



Brother, these images in the hoarding are so big! How do they make such big hoardings?

What's a vector image?
How do I make one?

Hoardings use vector images,
so that they don't look blurry
when they're made larger.

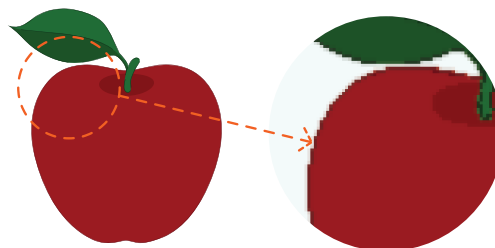
Here, let's make one now.



An image is a visual representation of information. A digital image is made up of **Pixels**. Pixels are the tiny boxes that come together to make up an image. This type of image is also called a **Raster image**.

Raster image

A raster image is an image file made up of pixels. Pictures taken with a digital camera are raster images. Raster images have one big disadvantage. They cannot be resized without becoming blurry.



Raster image

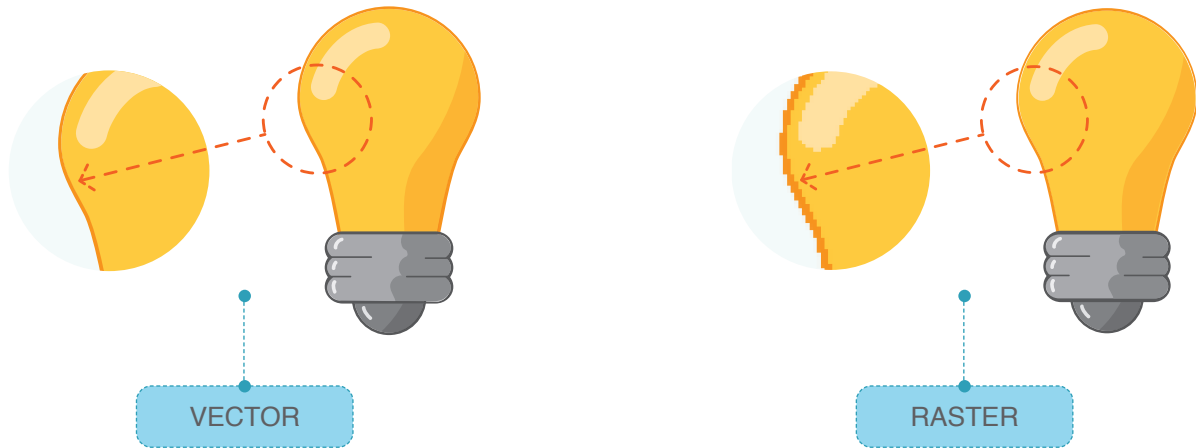
Since a raster image is made up of pixels, there is no visible loss of quality when it is resized to a smaller size. This is because the pixels become even less visible. On the other hand, when we resize an image to make it larger, the pixels become enlarged and more visible. This makes the image look blurry and pixelated.

Vector images are another type of image. They retain their appearance regardless of size.

Vector image

Vector images can be resized as much as required, without any loss of quality. Instead of pixels, they are made up of points, lines, and curves that are based upon mathematical formulas.

Raster images	Vector images
Made up of pixels, each containing information of a specific colour.	Made up of points, lines and curves, based on mathematical formula.
Resolution dependent – difficult to resize and edit without loss of quality.	Resolution independent – can be resized to any size without loss of quality.
Take up more memory and storage space.	Take up less memory and storage space.
Used for photographs, images, web graphics.	Used for logos, illustrations, signages.
Ideal for digital devices.	Ideal for prints.



Difference between Vector and Raster images

Exercise

Fill in the blanks

- 1 Computer graphics that are made up of points, lines, curves and are based upon mathematical formulas are called _____.
- 2 _____ graphics take up more memory and storage space than _____ graphics.

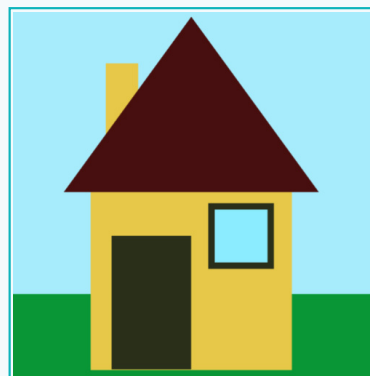
In this lesson, we will learn to create vector images.

Bytes

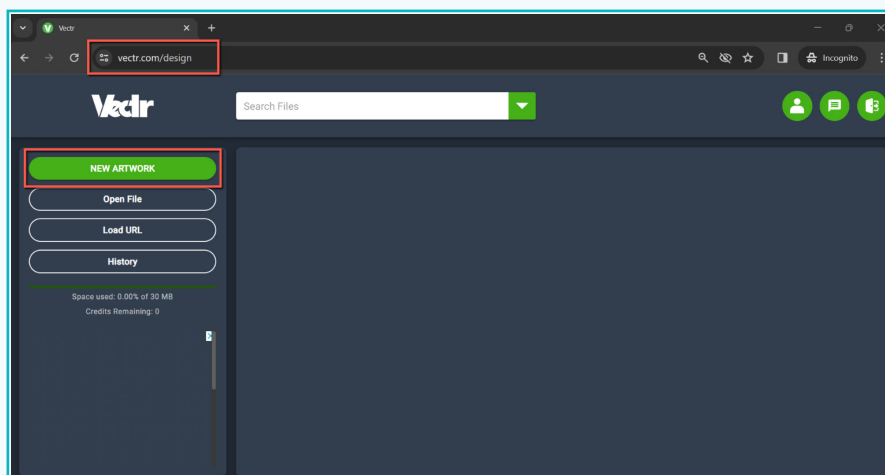
Graphics are visual images on a surface. It could be a wall, canvas, screen, paper, or stone. The purpose of graphics is to inform, illustrate, or entertain.

Activity

Create a vector image of a house.

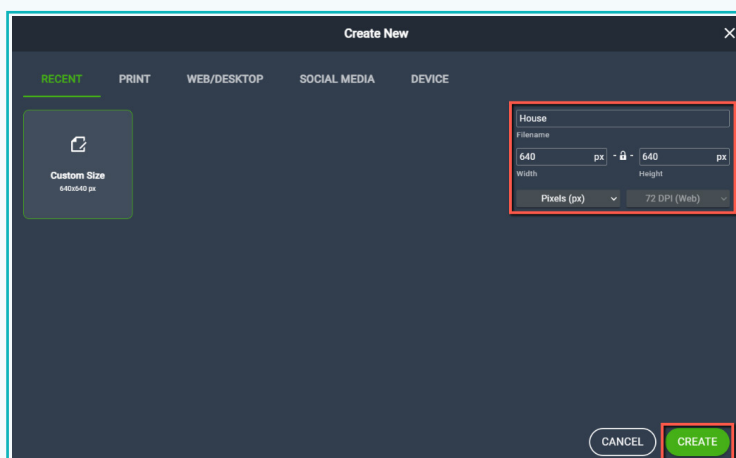


1 Load vectr.com on the browser.



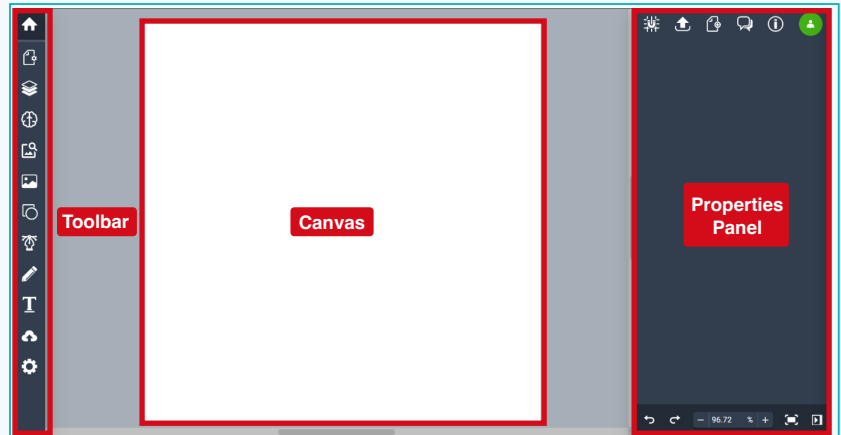
2 Set the filename as House. Click on Create.

We will use three important parts of the interface for this activity. These are the Toolbar, the Canvas and the Properties Panel.

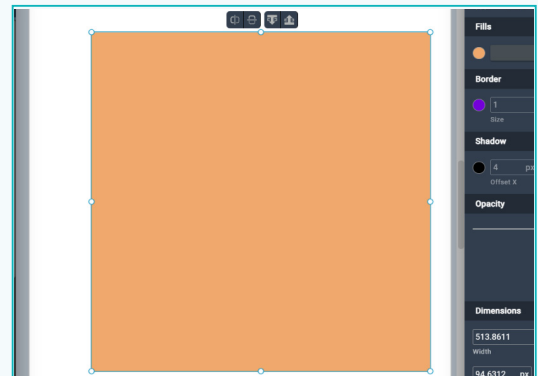
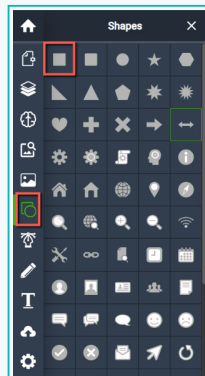


Activity

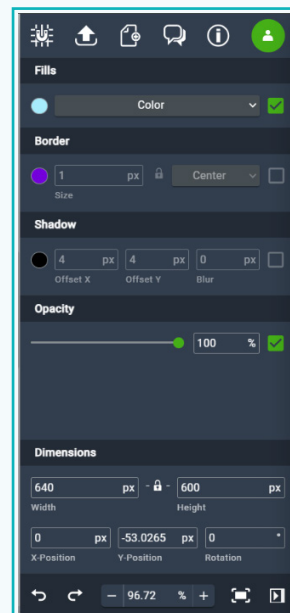
The canvas is the area where we design an artwork. The toolbar consists of tools that are used to create an artwork. The properties Panel shows the various properties of a specific tool and options to export a vector file.



- a. Select the rectangle tool. Click and drag on the canvas to draw a rectangle.

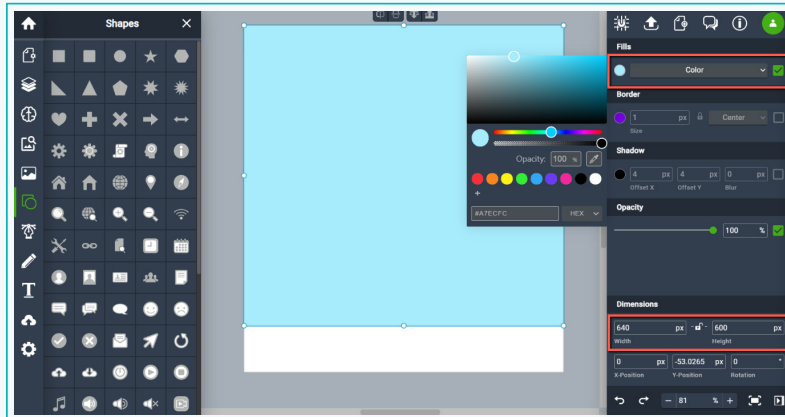


- b. To the right of the canvas is the properties bar. Here, we can set various properties for the image we create, such as color, border color, and shadows.

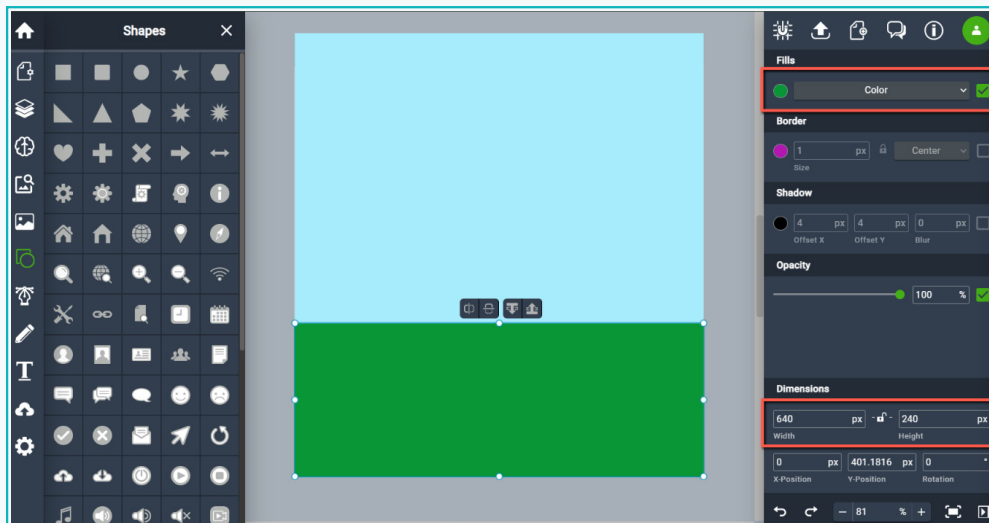


Activity

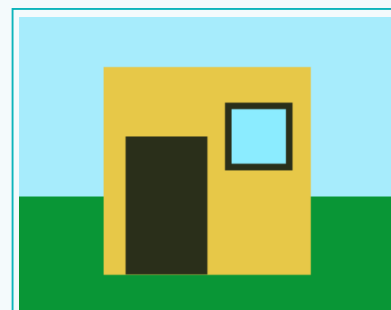
- c. For now, we will change the colour of the rectangle. As shown in the image below, click on the color option and select a shade of blue to use as the sky.



- d. Similarly, create a rectangle for the ground and set its colour to green.

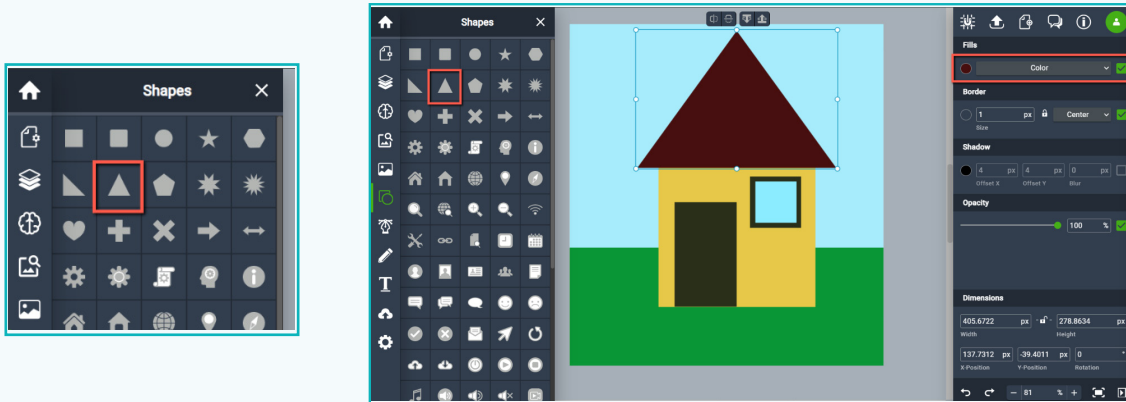


- e. Repeat these steps to create other elements of the clipart, such as the door and window.

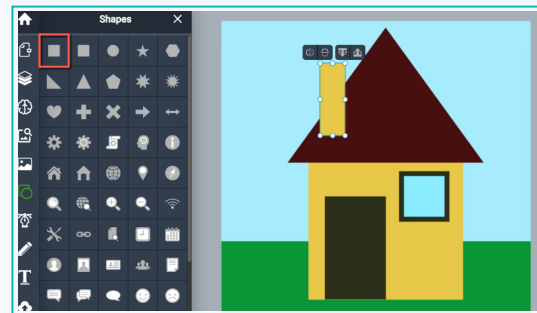


Activity

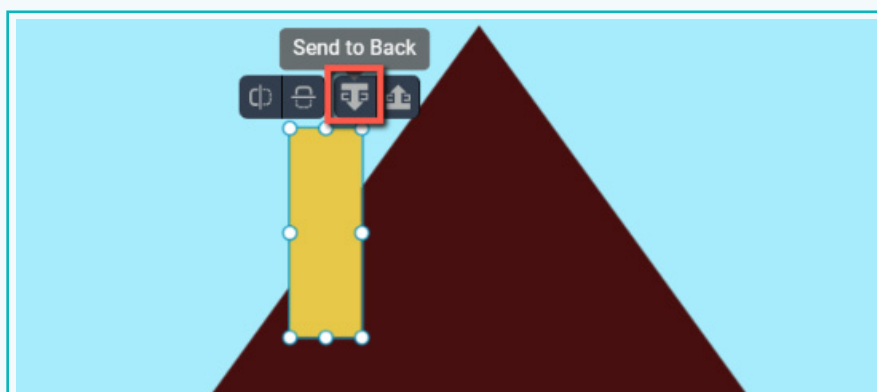
f. For the roof, we will use a triangle.



g. We will use a rectangle to add a chimney to the house.

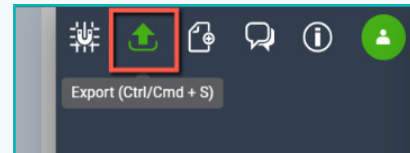


h. When we create the rectangle, it overlaps with the roof. We need it to be under the triangle. Select the rectangle and select the 'Send to Back' option.



Activity

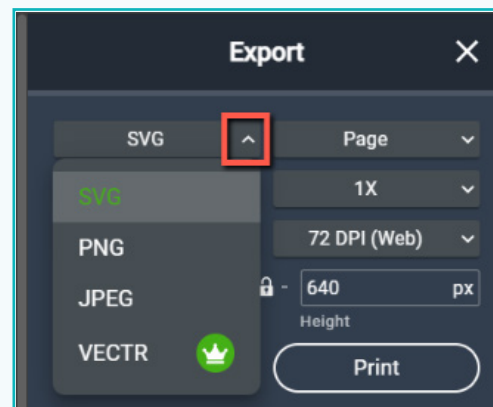
- i. The artwork is now complete! Now, we need to save it. Click on export.



- j. A window opens with an export panel. This panel lets us choose the image format to export in.

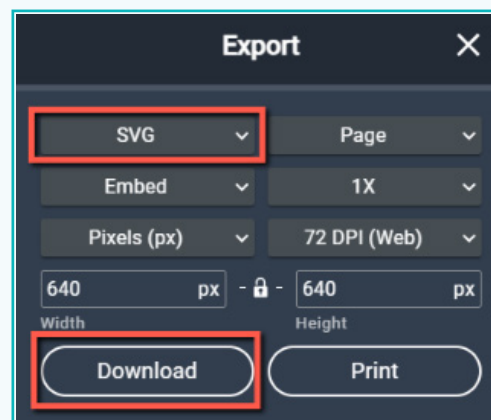
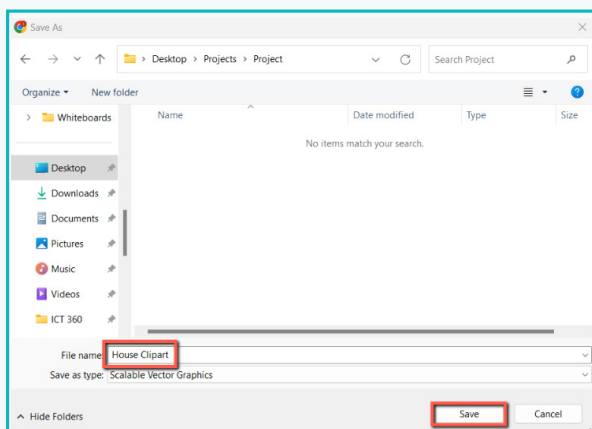
There are 3 formats to choose from:

- a. **JPG** - JPG is a file extension for a lossy raster graphics file.
- b. **PNG** - PNG is a raster graphics file format which supports lossless data compression.
- c. **SVG** - SVG is a vector graphic format used to display a variety of graphics on the web and in other places.



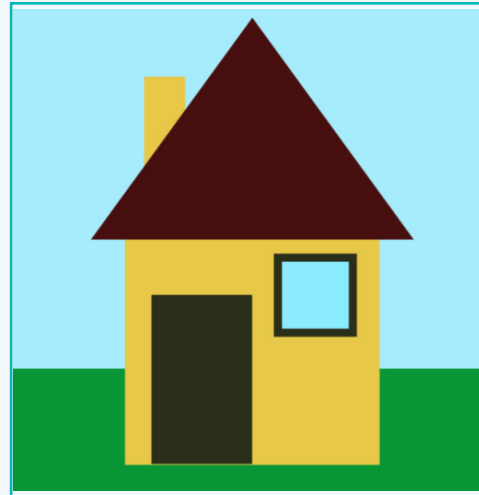
3 Save the file.

- a. Select SVG file format.
- b. Click on downloads to download the file and save it on the computer. Name it house clipart.



Activity

 Final output



Exercise

 Fill in the blanks

- 3 The _____ is the area where we design an artwork.
- 4 The _____ consists of tools we use to create an artwork.
- 5 The _____ command is used to move an object behind another object
- 6 The _____ panel is used to edit the properties of a shape.
- 7 When a raster image is scaled up, it usually _____ quality.

When a raster image is resized, it loses its quality and looks blurry. This is because the pixel information is changed. Since vector images are not made up of pixels, they can be stretched to any size without any loss of quality. The smooth lines and shapes in vector images produce high quality printed materials, with consistent colour and clear text.



Tech Flash

- **Vector graphic** : A vector graphic is made up of points, lines, and curves that are based upon mathematical formulas.
- **SVG** : SVG is a vector graphic format used to display a variety of graphics on the web and other places.
- **Raster graphic** : A raster graphic is an image file made up of pixels.